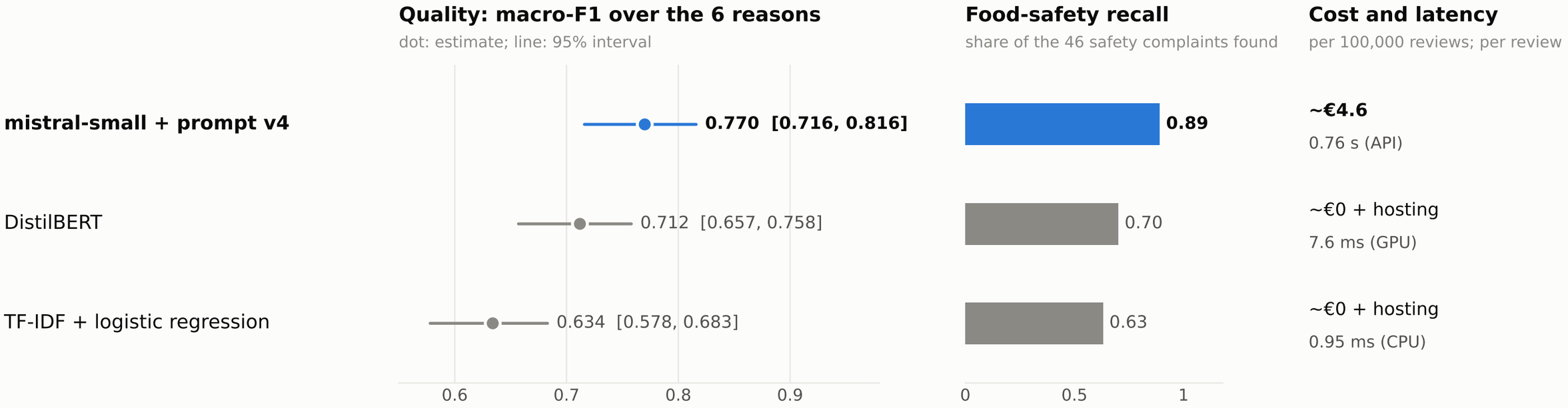
# Sorting negative customer reviews by complaint reason: which model?

Amazon Grocery reviews rated 1 to 3 stars, 6 reasons (a review can carry several).
Test: 300 randomly drawn reviews, hand-annotated before any prediction; 95% bootstrap confidence intervals.




## Recommendation: mistral-small (pinned version) with prompt v4, for batch sorting

- Best of the three on every quality metric; against DistilBERT, the gap is demonstrated on exact match (0.763 vs 0.683).
- Macro-F1 gap with DistilBERT: −0.058 [−0.115, +0.003], not demonstrated on 300 reviews.
- Food safety: recall 0.89 vs 0.70 (gap +0.20 [+0.07, +0.34]), ~3,000 more complaints caught per 100,000 reviews.
- Actual cost ~€4.6 per 100,000 reviews: prompt caching cuts the list price ($14.6) by 2.7×.
- Conditions: pinned model version; 100 reviews annotated each month to track quality; DistilBERT as a fallback.
- Switch to DistilBERT if: data must stay in-house, real-time is required, or ~1 million reviews per month.

In-house models trained on 4,979 reviews labelled by the LLM (distillation). DistilBERT latency on a T4 GPU. All decisions and figures: docs/CONVENTIONS.md (French).
Safety label singled out after the TF-IDF test run; with a Bonferroni correction over the 6 labels, the recall gap stays positive: [+0.02, +0.38].